THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 05

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** climate patterns	Climate change
2	**(b)** Checking claims with observation	Evidence-based thinking
3	**(d)** Lactobacillus – curd	Roles of microbes
4	**(d)** preventive	Used before the disease
5	**(a)** recover and limit spread	Good practice
6	**(c)** the current increases	More current gives more heat
7	**(d)** electrodes	Dipped in electrolyte
8	**(b)** two interacting objects	Forces are interactions
9	**(d)** Assam	Pre-monsoon
10	**(d)** the area is small so pressure is large	Same force on small area
11	**(a)** interparticle forces of attraction	Weaker or stronger by state
12	**(d)** Distilled water	Single substance
13	**(b)** Oxygen	Cannot be broken down
14	**(b)** arrange to take up more space	Expansion on freezing
15	**(a)** converge	Converging lens
16	**(b)** equinoxes	Around 21 March and 23 September
17	**(a)** clean air, water, food and climate regulation	Benefits from nature
18	**(b)** aquatic ecosystem	Water-based
19	**(b)** temperature is not too extreme during the year	Constant distance from Sun
20	**(c)** using natural resources sustainably	Community role

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Bacteria that break down crude oil (1). They help clean up oil spills (1).	1+1
22	An old Indian method of protecting against smallpox (1). Material from a sore was used to cause mild infection and immunity (1).	1+1
23	Helps animals navigate (1). Shields Earth from harmful space particles (1).	1+1
24	Air moves from high pressure to low pressure region (1). Flow continues until pressures equalise (1).	1+1
25	Ancient mixture of two or more metals (1). Had properties distinct from the metals (1).	1+1
26	Volume = mass ÷ density (1). 250 cm^3 (1).	1+1
27	Non-native species that spread widely (1). They harm native species (1).	1+1
28	Reduce, reuse and recycle (1). Use clean energy and public transport (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	It shows the same method of observation helps (1). Humans influence climate (1). Science can guide actions to protect Earth (1).	1+1+1
30	Lifting a bag (1). Chewing food (1). Heart pumping blood (1).	1+1+1
31	Particles move randomly (1). They collide with walls (1). Collisions produce pressure (1).	1+1+1
32	Left–right reversal in the image (1½). Writing appears reversed, e.g. ‘AMBULANCE’ is written reversed so it reads correctly in a rear-view mirror (1½).	1½+1½
33	Planning agriculture (1). Fixing festivals (1). Keeping records of events (1).	1+1+1
34	Delicate balance of climate (1). Changes affect weather and ecosystems (1). Could endanger species (1).	1+1+1

35	Zinc container is the negative terminal (1½). Carbon rod with metal cap is the positive terminal (1½).	1½+1½
	OR: Bring compass near an end (1). Needle's north-seeking end points to the south pole of the coil (1). Opposite end is the other pole (1).	1+1+1
36	Plant trees (1). Reduce pollution and plastic (1). Save water and energy (1).	1+1+1
	OR: Energy is lost at each level (1½). Less energy supports fewer organisms (1½).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	Four diseases with causal agent (2). Symptoms (1½). Preventive measures (1½).	2+1½+1½
	OR: Misuse lets resistant bacteria survive (1). They multiply (1). Spread through animals, soil, food and hospitals (1). Four controls: use only when prescribed, complete dose, avoid unnecessary use, do not give to cattle without advice (2).	1+1+1+2
38	Diagram (1). Metal rod above the building (1). Other end in ground (1). Provides easy path (1). Charges flow harmlessly to ground (1).	1+1+1+1+1
	OR: Rubbing ice and water charges clouds (1). Top positive, bottom negative (1). Air insulation breaks, flash (1). Safety: stay low, avoid tall objects (1). Avoid water and metals (1).	1+1+1+1+1
39	(i) 3 g/cm³ (2). (ii) Sinks since density is more than water (1). (iii) 150 g (2).	2+1+2
	OR: Solute and solvent (1). Solution (1). Saturated (1). Unsaturated (1). Solubility (1).	1+1+1+1+1
40	Pests resistant (1). Pollinators decline (1). Biodiversity falls (1). Organic farming (1). Crop rotation and manure (1).	1+1+1+1+1
	OR: Trophic levels (1). Producers at base (1). Consumers above (1). Energy decreases upward (1). Fewer organisms at top (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER KEY